Exercise semester 1/2569

Exercise 2 Schedule management (critical path method) (10 points)

Submit your answer in a **pdf file**. Name the file by **your name and student ID**.

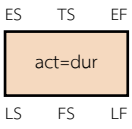
The table below shows the network diagram data of a project. All estimated durations are in days.

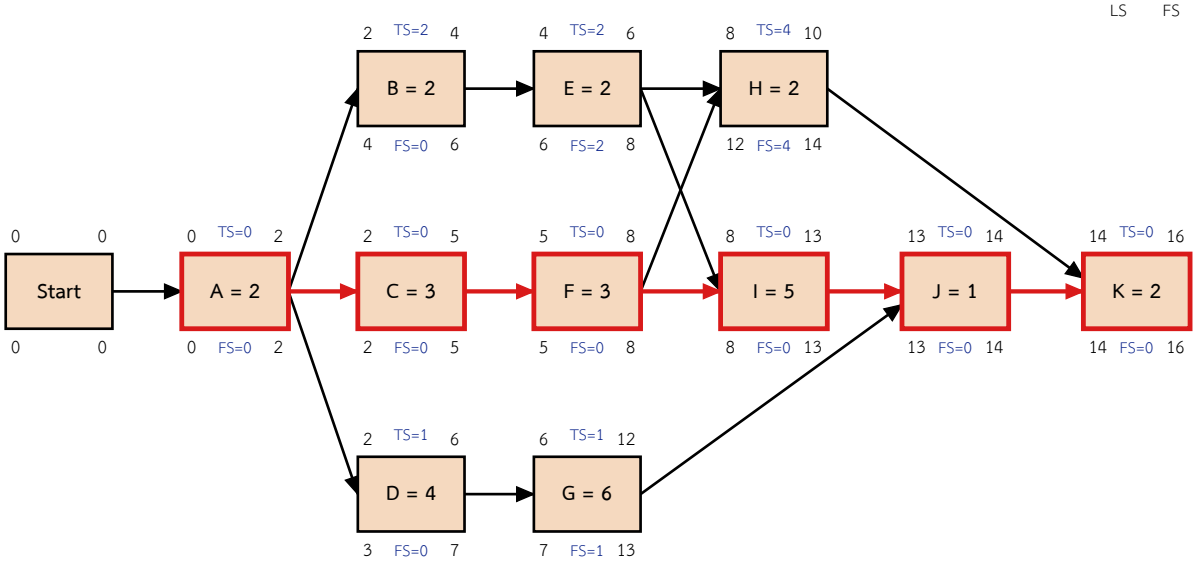
Activity	Predecessor	Estimated duration
A	-	2
B	A	2
C	A	3
D	A	4
E	B	2
F	C	3
G	D	6
H	E, F	2
I	E, F	5
J	G, I	1
K	H, J	2

1. Draw a PDM (activity on node) network diagram of the project and calculate the ES, EF, LS, LF, TS, FS of each activity (similar to slide 33 of chapter 6).

Forward pass: ES = max EF of predecessors, EF = ES + dur. Backward pass: LF = min LS of successors, LS = LF – dur. TS = LS – ES; FS = (min ES of successors) – EF. Critical path A-C-F-I-J-K, project duration 16 days.

Legend: ES = early start, EF = early finish, LS = late start, LF = late finish, TS = total slack, FS = free slack (days). Red = critical path.





2. Identify all paths on the network diagram and their lengths.

Path 1: A-B-E-H-K length = 2+2+2+2+2 = 10 days
Path 2: A-B-E-I-J-K length = 2+2+2+5+1+2 = 14 days
Path 3: A-C-F-H-K length = 2+3+3+2+2 = 12 days
Path 4: A-C-F-I-J-K length = 2+3+3+5+1+2 = 16 days (longest)
Path 5: A-D-G-J-K length = 2+4+6+1+2 = 15 days

3. The critical path for this project is

A-C-F-I-J-K (Path 4) – the longest path through the network diagram; every activity on it has TS = FS = 0.

It takes 16 days.

4. The shortest possible time needed to complete this project is 16 days.

5. Which activities may be delayed and their delay will not affect the immediately following activities? Explain.

E, G and H – the activities with free slack FS > 0 (E = 2, G = 1, H = 4 days); every other activity has FS = 0.

FS = (min ES of successors) – EF: E may finish as late as day 8 (H and I also wait for F), G day 13 (J waits for I), H day 14 (K waits for J).

6. Which activities may be delayed and their delay will not affect the project finish date? Explain.

B, D, E, G and H – the activities with total slack TS > 0 (B = 2, D = 1, E = 2, G = 1, H = 4 days).

TS = LS – ES: a delay within the TS leaves the project finish date (day 16) unchanged; A, C, F, I, J, K are critical (TS = 0).

7. If B is delayed and starts on day 5, choose one.

☑ The project finish date is delayed for at least 1 day(s).

☐ The project finish date is not affected by this delay.

Explain your choice.

B has ES = 2, LS = 4, TS = 2 days (day numbers on the same scale as ES/LS: B normally starts on day 2). Starting on day 5 delays B by 3 days, 1 day more than its total slack: B 5–7, E 7–9, I 9–14, J 14–15, K 15–17. The project finishes on day 17 instead of 16, at least 1 day late (A-B-E-I-J-K becomes the critical path).

8. If H is delayed and starts on day 10, choose one.

☐ The project finish date is delayed for at least ..... day(s).

☑ The project finish date is not affected by this delay.

Explain your choice.

H has ES = 8, LS = 12, TS = 4 days (FS = 4). Starting on day 10 delays H by 2 days, within its total slack: H finishes on day 12, still before the early start of K on day 14 (K must also wait for J, EF = 14), so K and the project finish date (day 16) are unchanged.

9. If D is delayed and starts on day 3, choose one.

☐ This delay affects G and J.

☐ This delay does not affect G and J.

☑ This delay affects G but does not affect J.

☐ This delay does not affect G but affects J.

Explain your choice.

D has ES = 2, LS = 3, TS = 1, FS = 0. Starting on day 3 delays D by 1 day: D finishes on day 7, so the early start of G moves from day 6 to 7 – G is affected (G 7–13, its slack is used up). J waits for G (EF 13) and I (EF 13), so its early start stays day 13: J and the project finish date (day 16) are not affected.